Block practice · Series and the binomial expansion

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Getting terms out of a formula for the sum

Series made of logarithms

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Practice workbook · Getting terms out of a formula for the sum

Getting the first term, common difference or common ratio, or a particular sum, directly from a formula for S_n or a sigma expression.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-02

1. Question 6 · part a · 4 marks

2020 NOV · Paper 01

Worked-example retry · Attempt part a only.

6 An arithmetic series A has first term a and common difference d.

The sum S_n of the first n terms of A is given by $S_n = n(15 + 2n)$

(a) Find the value of a and the value of d.

(4)

(b) Find the 20th term of A.

(2)

Given that $S_{2p} - 2S_p = 1 + S_{(p-1)}$

(c) find the value of p.

(4)

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1. Question 6 · part a · 4 marks · Working and answer

Working and answer

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2. Question 6 · part a · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part a only.

6 The sum of the first n terms of an arithmetic series is S_n where $S_n = n(3 + 2n)$

(a) Find the value of S_{20}

(2)

Given that $S_n = \sum_{r=1}^{n} (Ar + B)$

(b) find the value of A and the value of B

(6)

A different arithmetic series has first term 7 and common difference 4

The sum of the first n terms of this series is T_n

(c) Use algebra to find the value of n for which $T_n = S_n + 252$

(5)

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2. Question 6 · part a · 2 marks · Working and answer

Working and answer

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3. Question 7 · part a.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part a.i only.

7 The sum of the first n terms of an arithmetic series is A_n where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference.

(2)

The sum of the first n terms of a geometric series is G_n where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio.

(2)

(c) Find the value of n for which $A_{14} - 6 = G_n$

(5)

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3. Question 7 · part a.i · 0 marks · Working and answer

Working and answer

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4. Question 7 · part a.ii · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part a.ii only.

7 The sum of the first n terms of an arithmetic series is A_n where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference.

(2)

The sum of the first n terms of a geometric series is G_n where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio.

(2)

(c) Find the value of n for which $A_{14} - 6 = G_n$

(5)

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4. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 7 · part b.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part b.i only.

7 The sum of the first n terms of an arithmetic series is A_n where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference.

(2)

The sum of the first n terms of a geometric series is G_n where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio.

(2)

(c) Find the value of n for which $A_{14} - 6 = G_n$

(5)

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5. Question 7 · part b.i · 0 marks · Working and answer

Working and answer

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6. Question 7 · part b.ii · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b.ii only.

7 The sum of the first n terms of an arithmetic series is A_n where

$$A_n = \sum_{r=1}^{n} (4r + 5)$$

(a) For this arithmetic series, find

(i) the first term,

(ii) the common difference. **(2)**

The sum of the first n terms of a geometric series is G_n where

$$G_n = \sum_{r=1}^{n} 4(3)^{r-1}$$

(b) For this geometric series, find

(i) the first term,

(ii) the common ratio. **(2)**

(c) Find the value of n for which $A_{14} - 6 = G_n$ **(5)**

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6. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Series made of logarithms

Series where terms are logarithms (or are turned into logarithms), so log laws link a geometric pattern to an arithmetic one.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-15

1. Question 4 · part whole · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part whole only.

4 The nth term of an arithmetic series is u_n where

$$u_n = (n + 1)\ln 4$$

Given that the sum of the first n terms of the series is S_n

show that $S_n = \ln 2^{(n^2+an)}$ where a is an integer whose value is to be found.

(5)

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1. Question 4 · part whole · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 5 marks)

1. Question 4 · part whole · 5 marks · Working and answer

Working and answer

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2. Question 3 · part whole · 7 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part whole only.

3 The nth term of an arithmetic series is u_n such that

$$u_n = \ln a + (n - 1)\ln b$$

where a and b are positive integers.

Given that $u_2 = \ln 12$ and that $u_5 = \ln 768$

find the value of a and the value of b.

(7)

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2. Question 3 · part whole · 7 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 7 marks)

2. Question 3 · part whole · 7 marks · Working and answer

Working and answer

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3. Question 5 · part c · 5 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

5 A geometric series G has first term 12 and common ratio $\frac{3}{8}$

(a) Find the sum to infinity of G

(2)

(b) Show that the 6th term of G can be written as $\frac{3^6}{2^{13}}$

(3)

The nth term of G is u_n

(c) By finding an expression for u_n in terms of n, show that

$$\log_2 u_n = n\log_2 3 - 3n + 5$$

(5)

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3. Question 5 · part c · 5 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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3. Question 5 · part c · 5 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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14

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3. Question 5 · part c · 5 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 10 marks)

3. Question 5 · part c · 5 marks · Working and answer

Working and answer

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